Geometry : Vectors — problem 12.1.28

A. Engel — Problem-Solving Strategies · 1998

The following problems except 40 and 41 are to be solved by complex numbers. Sometimes a convenient choice of the origin is helpful.

A triangle with vertices a, b, c is equilateral iff $a^2 + b^2 + c^2 - ab - bc - ca = 0$.